

SMALL INNOVATIVE LAUNCHER FOR EUROPE

LAUNCHER SYSTEM CONCEPT DESIGN WP1



LAUNCHER SYSTEM REQUIREMENTS DOCUMENT (D1.4)

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Publishable Summary

Today's market for small satellites is expanding, but there is little capacity for affordable launches. For a dedicated small satellite launch vehicle, cost reduction is essential to compete with ride-shares, e.g. through reuse, low cost components, volume production, optimised manufacturing. The goal of this document is to define the top level requirements for a small satellite launch vehicle. As the main objective is to create an economically feasible launcher, a certain level of flexibility regarding the requirements is needed.

Up to now microsatellites almost exclusively have been launched as a secondary payload. The use case for a small satellite launch vehicle is different from that of the conventional counterparts. For instance, the launch rate needs to be sufficiently high to be economically feasible, which impacts the launch campaign. Typically, microsatellite customers will contract a launch between 1 and 3 years before launch, whereas CubeSat customers are seeking launches between 6 months and 1 year.

In a conventional launch, most technical parameters are dictated by the primary payload, and secondary payloads usually do not have a say in many aspects of the launch. For a dedicated small satellite launch vehicle, customer expectations will be higher; hence the launch vehicle should be based on a set of technical requirements on orbit, interfaces, and environment that will satisfy the majority of small satellite missions.

Regulations for launching satellites into orbit include the topics of safety, reliability, and debris mitigation. Currently, there are no European regulations on orbital launches, except for a reference to a generic ISO norm (ISO 14620-1:2002). Until European legislation is in place, it is up to the launching nation to define the regulations. At the moment, Norway - as the anticipated launching state - only has regulations for sub-orbital flights with sounding rockets. This should in the future be extended towards orbital launches, which is a political decisions involving many aspects, including liability of the nation and insurance against accidents.

In the USA, the Federal Aviation Authority (FAA) has defined a safety requirement for launchers on casualty expectation. Human casualty is defined as a fatality or serious injury from a piece of vehicle debris. Risk is defined as the product of the probability of occurrence of an event and the consequences of that event. The FAA has set an acceptable limit for the risk of expected casualty to 0.00003 per launch.

Prior to the launch, the range safety officers determine how far, and in what direction, the rocket is allowed to deviate from the planned trajectory, so that unsafe situations are avoided if a malfunction should happen. During its ascent, the rocket is monitored using both radar and on-board telemetry systems. In case of a safety-related event, the launch should be terminated in a safe way, for example by a flight termination system.



Redundancy can lead to a more reliable system, so redundancy increases both reliability and safety. However, it also has a cost impact, as it results in a more complex design (hot/cold/standby, power lines, cross-strapping, duplicate interfaces), which is more expensive (procurement and testing) and also has a performance penalty (mass increase). Still, redundancy is an option, especially since sensors and computers are becoming smaller and lighter. Absolute numbers do make a difference in that a redundant computer of 20 kg has a much bigger impact than one that weighs only 100 grams.

Regarding orbital debris, the market prospect for small satellites not only leads to many initiatives for a small launcher, but also to many satellites in orbit. With the anticipated launch rate and a fully loaded launcher, some 200 CubeSats will be put into orbit each year. This may lead to further regulation of the space environment, for instance by limiting the lifetime for In-Orbit Demonstration (IOD) satellites to a few years or even less which sets a limit to the maximum altitude for such IOD launches.

The main requirements for a small satellite launch vehicle are defined following the use case for a typical (microsatellite or CubeSat) customer. The most important requirement is to have a safe launch with a casualty expectation. Most other system requirements are about the (interface with the) payload. The requirements on the payload capacity are based on a cost-benefit analysis in the Business Development Plan. Some requirements are flown down to the subsystem level (Engines, Structures, and Avionics), namely those dealing with safety, environment, and the low cost objective.